Knowledge Representation and Machine Learning Basics

Artificial Intelligence - Units 3-4 - Reasoning and learning

Propositional and predicate logic

- Propositional logic uses statements joined by AND, OR, NOT, implies.
- First-order logic adds objects, predicates and quantifiers: forall x Student(x) -> Studies(x).
- Inference methods include modus ponens and resolution.

Machine learning in three types

- Supervised: learn from labelled examples (classification, regression).
- Unsupervised: find structure in unlabelled data (clustering, dimensionality reduction).
- Reinforcement: learn actions through rewards and penalties.

A typical supervised workflow

```
1. Collect and clean data
2. Split into train / validation / test sets
3. Pick a model (e.g. decision tree, logistic regression)
4. Train on the training set, tune on validation
5. Report final metrics once on the test set
```

Overfitting

A model that memorizes training noise performs well on training data and poorly on new data. Remedies: more data, simpler models, regularization, early stopping and cross-validation.

Evaluating classifiers

```
accuracy  = (TP + TN) / (TP + TN + FP + FN)
precision = TP / (TP + FP)
recall    = TP / (TP + FN)
F1        = 2 * precision * recall / (precision + recall)
```

Worked example: TP=40, FP=10, FN=20, TN=30 -> precision 0.80, recall 0.667, accuracy 0.70.

Ethics

Responsible AI considers bias in data, transparency, privacy and accountability for automated decisions.